



△ **Platinum ring** set with yellow and white diamonds

Diamond

With exceptional beauty, lustre, and sparkle, diamond is the most iconic of all precious stones and highly prized in jewellery all over the world. However, this is only one of its uses. Industrial diamond is a vital component in oil drilling, specialist scalpels, tool manufacturing, and many other industries, all of which use the supreme hardness of diamonds for cutting tools and abrasive powders. There is no firm boundary between gem-grade and industrial-grade diamonds – around 80 per cent of the diamonds mined each year are unsuitable for gemstone wear, and find other uses in industry. However, very small or lower-grade stones can be polished into gemstones rather than being used in industry.

Discovering diamonds

For over 2,000 years, diamonds were found only as crystals in river gravels, and, until 1725, India was the major source. As Indian production waned, diamonds were discovered in Brazil, and in 1867 they were found in gravels near the Orange River in the Kimberley region of South Africa. Further exploration there revealed volcanic pipes of a previously unknown rock type containing diamonds; this was named kimberlite and was recognized as the diamond source rock. Its discovery formed the basis of the modern diamond industry. Many similar pipes have since been found in other African countries, Siberia, Australia, and more recently in Canada, China, and the USA.

Key pieces



Dresden Green

Dresden Green | Probably from the Kollur Mine in India, this 41-carat natural green diamond is named after its home in Dresden, Germany (see pp.140–41). It is famous for its extraordinary green colour and is set in a lavish hat ornament.

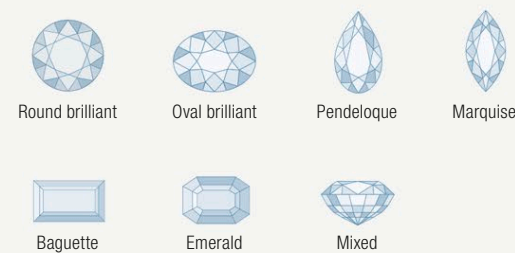


Cullinan I

Cullinan I | Originally part of a 3,106.75-carat rough stone from which several gems were cut (see p.54), the Cullinan I is part of the British crown jewels. At 530.1 carats, it is the largest polished white diamond in the world.

Specification

Chemical name Carbon | **Formula** C | **Colours**
All colours | **Structure** Cubic | **Hardness** 10
SG 3.4–3.5 | **RI** 2.42 | **Lustre** Adamantine
Streak None



Locations

1 Canada 2 USA 3 Brazil 4 Ghana 5 Angola 6 Namibia
7 Botswana 8 South Africa 9 India 10 Russia 11 Borneo
12 Australia



Platinum setting

Allnatt Diamond | Weighing 101.29 carats, this extraordinary stone is described as a fancy vivid yellow and is set in a platinum flower design. It is named after one of its former owners, businessman and art collector Alfred Ernest Allnatt.